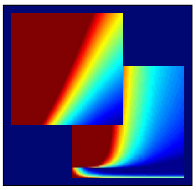
CS/CNS/EE 156a Learning Systems
Caltech - Fall 2025
https://caltech.instructure.com/courses/8882
(Learning From Data campus version)



Homework # 4

Due Monday, October 27, 2025, at 2:00 PM PDT

*Definitions and notation follow the lectures. All questions have multiple-choice answers (***[a], [b], [c], ...***). Collaboration is allowed but without discussing selected or excluded choices. Your solutions must be based on your own work. See the initial "***Course Description and Policies***" handout for important details about collaboration, open book, and chatGPT policies.*

Note about the homework

- Answer each question by deriving the answer (carries 6 points) then selecting from the multiple-choice answers (carries 4 points). You can select 1 or 2 of the multiple-choice answers for each question, but you will get 4 or 2 points, respectively, for a correct answer. See the initial "**Course Description and Policies**" handout for important details.
- The problems range from easy to difficult, and from practical to theoretical. Some problems require running a full experiment to arrive at the answer.
- The answer may not be obvious or numerically close to one of the choices, but one (and only one) choice will be correct if you follow the instructions precisely in each problem. You are encouraged to explore the problem further by experimenting with variations on these instructions, for the learning benefit.
- You are encouraged to take part in the Piazza discussion forum. Please make sure you don't discuss specific answers, or specific excluded answers, before the homework is due.

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- **Generalization Error**

In Problems 1-3, we look at generalization bounds numerically. For $N > d_{\text{VC}}$, use the simple approximate bound $N^{d_{\text{VC}}}$ for the growth function $m_{\mathcal{H}}(N)$.

1. For an $\mathcal{H}$ with $d_{\text{VC}} = 10$, if you want 95% confidence that your generalization error is at most 0.05, what is the closest numerical approximation of the sample size that the VC generalization bound predicts?

[a] 400,000 We can use the formula from lecture and plug in d_VC=10, ϵ=0.05, δ=0.05

[b] 420,000 This yields m=429040.28

[c] 440,000

[d] 460,000

[e] 480,000

2. There are a number of bounds on the generalization error ϵ, all holding with probability at least $1-\delta$. Fix $d_{\text{VC}} = 50$ and $\delta = 0.05$ and plot these bounds as a function of N. Which bound is the smallest for very large N, say $N = 10,000$? Note that [c] and [d] are implicit bounds in ϵ. Plug in: d_VC=50, δ=0.05, N=10,000

[a] Original VC bound: $\epsilon \leq \sqrt{\frac{8}{N} \ln \frac{4m_{\mathcal{H}}(2N)}{\delta}}$ ϵ≈0.632; m_H(2N)=(2*10000)^50

[b] Rademacher Penalty Bound: $\epsilon \leq \sqrt{\frac{2\ln(2Nm_{\mathcal{H}}(N))}{N}} + \sqrt{\frac{2}{N}\ln\frac{1}{\delta}} + \frac{1}{N}$ ϵ≈0.331; m_H(N)=(10000)^50

[c] Parrondo and Van den Broek: $\epsilon \leq \sqrt{\frac{1}{N}(2\epsilon + \ln\frac{6m_{\mathcal{H}}(2N)}{\delta})}$ ϵ≈0.223; using CAS

[d] Devroye: $\epsilon \leq \sqrt{\frac{1}{2N}(4\epsilon(1+\epsilon) + \ln\frac{4m_{\mathcal{H}}(N^2)}{\delta})}$ ϵ≈0.215; using CAS

[e] They are all equal.

3. For the same values of d_{VC} and δ of Problem 2, but for small N, say $N = 5$, which bound is the smallest?

[a] Original VC bound: $\epsilon \leq \sqrt{\frac{8}{N} \ln \frac{4m_{\mathcal{H}}(2N)}{\delta}}$ ϵ≈15.71

[b] Rademacher Penalty Bound: $\epsilon \leq \sqrt{\frac{2\ln(2Nm_{\mathcal{H}}(N))}{N}} + \sqrt{\frac{2}{N}\ln\frac{1}{\delta}} + \frac{1}{N}$ ϵ≈8.15;

[c] Parrondo and Van den Broek: $\epsilon \leq \sqrt{\frac{1}{N}(2\epsilon + \ln\frac{6m_{\mathcal{H}}(2N)}{\delta})}$ ϵ≈5.76

[d] Devroye: $\epsilon \leq \sqrt{\frac{1}{2N}(4\epsilon(1+\epsilon) + \ln\frac{4m_{\mathcal{H}}(N^2)}{\delta})}$ ϵ≈6.12

[e] They are all equal.

- **Bias and Variance**

Consider the case where the target function $f : [-1, 1] \to \mathbb{R}$ is given by $f(x) = \sin(\pi x)$ and the input probability distribution is uniform on $[-1, 1]$. Assume that the training set has only two examples (picked independently), and that the learning algorithm produces the hypothesis that minimizes the mean squared error on the examples.

4. Assume the learning model consists of all hypotheses of the form $h(x) = ax$. What is the expected value, $\bar{g}(x)$, of the hypothesis produced by the learning algorithm (expected value with respect to the data set)? Express your $\bar{g}(x)$ as $\hat{a}x$, and round $\hat{a}$ to two decimal digits only, then match *exactly* to one of the following answers.

[a] $\bar{g}(x) = 0$

[b] $\bar{g}(x) = 0.79x$

[c] $\bar{g}(x) = 1.07x$

[d] $\bar{g}(x) = 1.58x$

[e] None of the above

See explanation on page 6-7 of this document

5. What is the closest value to the bias in this case?

[a] 0.1

[b] 0.3

0.271

[c] 0.5

See explanation on page 7 of this document

[d] 0.7

[e] 1.0

6. What is the closest value to the variance in this case?

[a] 0.2

[b] 0.4

0.218

See explanation on page 7-8 of this document

[c] 0.6

[d] 0.8

[e] 1.0

7. Now, let's change $\mathcal{H}$. Which of the following learning models has the least expected value of out-of-sample error? f(x) is odd and nonlinear but somewhat locally linear near origin

[a] Hypotheses of the form $h(x) = b$ constant model can't follow odd function high bias

[b] Hypotheses of the form $h(x) = ax$ odd linear model; low bias, medium variance

[c] Hypotheses of the form $h(x) = ax + b$ f(x) is centered around origin so [b] is better than [c]

[d] Hypotheses of the form $h(x) = ax^2$ quadratic good for even functions; high bias for f(x)

[e] Hypotheses of the form $h(x) = ax^2 + b$ quadratic good for even functions; high bias for f(x)

- **VC Dimension**

8. Let $q \geq 1$ be an integer, and assume that $m_{\mathcal{H}}(1) = 2$. What is the VC dimension of a hypothesis set whose growth function for all $N \geq 1$ satisfies: $m_{\mathcal{H}}(N+1) = 2m_{\mathcal{H}}(N) - \binom{N}{q}$? Recall that $\binom{M}{m} = 0$ when $m > M$.

[a] $q - 2$

[b] $q - 1$

[c] q

[d] $q + 1$

[e] None of the above

For small N < q, since C(N, q) = 0:
m_H(N+1)=2m_H(N)
Starting from m_H(1)=2, we get
m_H(N)=2^N for N ≤ q
This means the hypothesis can shatter q points.

For N = q;
m_H(q+1)=2m_H - C(q, q) = 2(2^q)-1
= 2^(q+1)-1 < 2^(q+1)
So once N=q+1, the growth function drops below 2^N. Therefore VC dim is q.

9. For hypothesis sets $\mathcal{H}_1, \mathcal{H}_2, ..., \mathcal{H}_K$ with finite, positive VC dimensions $d_{\text{VC}}(\mathcal{H}_k)$ (same input space $\mathcal{X}$), some of the following bounds are correct and some are not. Which, among the correct ones, is the tightest bound (the smallest range of values) on the VC dimension of the **intersection** of the sets: $d_{\text{VC}}(\bigcap_{k=1}^{K} \mathcal{H}_k)$? (The VC dimension of an empty set or a singleton set is taken as zero)

[a] $0 \leq d_{\text{VC}}(\bigcap_{k=1}^{K} \mathcal{H}_k) \leq \sum_{k=1}^{K} d_{\text{VC}}(\mathcal{H}_k)$

[b] $0 \leq d_{\text{VC}}(\bigcap_{k=1}^{K} \mathcal{H}_k) \leq \min\{d_{\text{VC}}(\mathcal{H}_k)\}_{k=1}^{K}$

[c] $0 \leq d_{\text{VC}}(\bigcap_{k=1}^{K} \mathcal{H}_k) \leq \max\{d_{\text{VC}}(\mathcal{H}_k)\}_{k=1}^{K}$

[d] $\min\{d_{\text{VC}}(\mathcal{H}_k)\}_{k=1}^{K} \leq d_{\text{VC}}(\bigcap_{k=1}^{K} \mathcal{H}_k) \leq \max\{d_{\text{VC}}(\mathcal{H}_k)\}_{k=1}^{K}$

[e] $\min\{d_{\text{VC}}(\mathcal{H}_k)\}_{k=1}^{K} \leq d_{\text{VC}}(\bigcap_{k=1}^{K} \mathcal{H}_k) \leq \sum_{k=1}^{K} d_{\text{VC}}(\mathcal{H}_k)$

H = ⋂_{k = 1}^{K} H_k

Because H is a subset of every H_k, it is also a subset of the least complex set. The intersection can only restrict the hypothesis; therefore:
0 ≤ d_VC(⋂_{k = 1}^{K} H_k) ≤ min_k d_VC(H_k)

Since H ⊆ H_min, H cannot be more powerful than H_min, it cannot shatter any set of points H_min can't shatter

10. For hypothesis sets $\mathcal{H}_1, \mathcal{H}_2, ..., \mathcal{H}_K$ with finite, positive VC dimensions $d_{\text{VC}}(\mathcal{H}_k)$ (same input space $\mathcal{X}$), some of the following bounds are correct and some are not. Which, among the correct ones, is the tightest bound (the smallest range of values) on the VC dimension of the **union** of the sets: $d_{\text{VC}}(\bigcup_{k=1}^{K} \mathcal{H}_k)$?

[a] $0 \leq d_{\text{VC}}(\bigcup_{k=1}^{K} \mathcal{H}_k) \leq \sum_{k=1}^{K} d_{\text{VC}}(\mathcal{H}_k)$

[b] $0 \leq d_{\text{VC}}(\bigcup_{k=1}^{K} \mathcal{H}_k) \leq K - 1 + \sum_{k=1}^{K} d_{\text{VC}}(\mathcal{H}_k)$

$$[\mathbf{c}]\ \ \min\{d_{\mathrm{VC}}(\mathcal{H}_k)\}_{k=1}^{K} \ \le\ d_{\mathrm{VC}}(\textstyle\bigcup_{k=1}^{K}\mathcal{H}_k) \ \le\ \sum_{k=1}^{K} d_{\mathrm{VC}}(\mathcal{H}_k)$$

$$[\mathbf{d}]\ \ \max\{d_{\mathrm{VC}}(\mathcal{H}_k)\}_{k=1}^{K} \ \le\ d_{\mathrm{VC}}(\textstyle\bigcup_{k=1}^{K}\mathcal{H}_k) \ \le\ \sum_{k=1}^{K} d_{\mathrm{VC}}(\mathcal{H}_k)$$

$$[\mathbf{e}]\ \ \max\{d_{\mathrm{VC}}(\mathcal{H}_k)\}_{k=1}^{K} \ \le\ d_{\mathrm{VC}}(\textstyle\bigcup_{k=1}^{K}\mathcal{H}_k) \ \le\ K-1+\sum_{k=1}^{K} d_{\mathrm{VC}}(\mathcal{H}_k)$$

Because H consists of every individual set, it also contains the most complex set H_max w/ d_VC = max{d_VC{H_k}}

Since H_max ⊆ H, H must be at least as powerful as H_max, it can shatter any set of points H_max can shatter.

Therefore d_VC(H) ≥ max{d_VC(H_K).

[d] has an upper bound of sum{d_K}, which deosn't make sense.

Counterexample:

The union of positive rays (d=1) and negative rays (d=1) created intervals (d=2) where d = sum{d_K}. But the union of two different sets of intervals does not create two intervals and d < sum{d_K}.

The question asks for the tightest correct bound. Since [d] is not a correct general bound, and [e] is, [e] is the only valid answer.

CS 156a PS4 Explanations

Om Sanan

27 October 2025

§4

Model: $h(x) = ax$. Target: $f(x) = \sin(\pi x)$, so $y_i = \sin(\pi x_i)$. Minimize Mean Squared Error, E_{in}:

$$E_{in}(a) = \frac{1}{2}\sum_{i=1}^{2}(h(x_i) - y_i)^2 = \frac{1}{2}\left[(ax_1 - y_1)^2 + (ax_2 - y_2)^2\right]$$

Find minimum by setting derivative w.r.t. a to 0:

$$\begin{aligned}
\frac{\partial E_{in}}{\partial a} &= (ax_1 - y_1)x_1 + (ax_2 - y_2)x_2 = 0 \\
ax_1^2 - y_1x_1 + ax_2^2 - y_2x_2 &= 0 \\
a(x_1^2 + x_2^2) &= y_1x_1 + y_2x_2 \\
a^{(\mathcal{D})} &= \frac{y_1x_1 + y_2x_2}{x_1^2 + x_2^2}
\end{aligned}$$

Substitute $y_i = \sin(\pi x_i)$:

$$a^{(\mathcal{D})} = \frac{x_1\sin(\pi x_1) + x_2\sin(\pi x_2)}{x_1^2 + x_2^2}$$

The expected hypothesis $\bar{g}(x)$ is:

$$\bar{g}(x) = \mathbb{E}_{\mathcal{D}}\left[a^{(\mathcal{D})}x\right] = \left(\mathbb{E}_{\mathcal{D}}\left[a^{(\mathcal{D})}\right]\right)x = \hat{a}x$$

We need to compute $\hat{a} = \mathbb{E}_{\mathcal{D}}\left[a^{(\mathcal{D})}\right]$. The input distribution is $P(x) = 1/2$ on $[-1, 1]$. $P(x_1, x_2) = P(x_1)P(x_2) = 1/4$.

$$\begin{aligned}
\hat{a} &= \mathbb{E}_{x_1,x_2}\left[\frac{x_1\sin(\pi x_1) + x_2\sin(\pi x_2)}{x_1^2 + x_2^2}\right] \\
\hat{a} &= \int_{-1}^{1}\int_{-1}^{1}\left(\frac{x_1\sin(\pi x_1) + x_2\sin(\pi x_2)}{x_1^2 + x_2^2}\right)P(x_1, x_2)\,dx_1\,dx_2 \\
\hat{a} &= \frac{1}{4}\int_{-1}^{1}\int_{-1}^{1}\frac{x_1\sin(\pi x_1) + x_2\sin(\pi x_2)}{x_1^2 + x_2^2}\,dx_1\,dx_2
\end{aligned}$$

By symmetry, the integral of the x_1 term is equal to the integral of the x_2 term:

$$\begin{aligned}
\hat{a} &= 2 \cdot \frac{1}{4}\int_{-1}^{1}\int_{-1}^{1}\frac{x_1\sin(\pi x_1)}{x_1^2 + x_2^2}\,dx_2\,dx_1 \\
\hat{a} &= \frac{1}{2}\int_{-1}^{1}x_1\sin(\pi x_1)\left(\int_{-1}^{1}\frac{1}{x_1^2 + x_2^2}\,dx_2\right)dx_1
\end{aligned}$$

Solvin the inner integral:

$$\int_{-1}^{1}\frac{1}{x_1^2 + x_2^2}\,dx_2 = \left[\frac{1}{x_1}\arctan\left(\frac{x_2}{x_1}\right)\right]_{x_2=-1}^{1}$$

$$= \frac{1}{x_1} \arctan\left(\frac{1}{x_1}\right) - \frac{1}{x_1} \arctan\left(\frac{-1}{x_1}\right) = \frac{2}{x_1} \arctan\left(\frac{1}{x_1}\right)$$

Substitute back:

$$\hat{a} = \frac{1}{2} \int_{-1}^{1} x_1 \sin(\pi x_1) \left(\frac{2}{x_1} \arctan\left(\frac{1}{x_1}\right)\right) dx_1$$
$$\hat{a} = \int_{-1}^{1} \sin(\pi x) \arctan\left(\frac{1}{x}\right) dx$$

Using CAS we get

$$\hat{a} \approx 1.428$$

§5

Bias over x:

$$\text{bias} = \mathbb{E}_x \left[(\bar{g}(x) - f(x))^2\right]$$

From (4), we have $\bar{g}(x) = \hat{a}x$ with $\hat{a} \approx 1.43$. The target is $f(x) = \sin(\pi x)$. The distribution is $P(x) = 1/2$ on $[-1, 1]$.

$$\text{bias} = \int_{-1}^{1} (\bar{g}(x) - f(x))^2 P(x)\, dx$$
$$\text{bias} = \int_{-1}^{1} (\hat{a}x - \sin(\pi x))^2 \cdot \frac{1}{2}\, dx$$

The integrand $k(x) = (\hat{a}x - \sin(\pi x))^2$ is an even function, since $\hat{a}x$ and $\sin(\pi x)$ are both odd functions, their difference is odd, and the square of an odd function is even. We can simplify the integral:

$$\text{bias} = 2 \cdot \int_{0}^{1} (\hat{a}x - \sin(\pi x))^2 \cdot \frac{1}{2}\, dx$$

$$\text{bias} \approx \int_{0}^{1} (1.43x - \sin(\pi x))^2\, dx$$

Using CAS we get:

$$\boxed{\text{bias} \approx 0.271}$$

§6

Variance of the hypothesis $g^{(\mathcal{D})}(x)$ with respect to the dataset $\mathcal{D}$:

$$\text{variance} = \mathbb{E}_x \left[\mathbb{E}_{\mathcal{D}} \left[(g^{(\mathcal{D})}(x) - \bar{g}(x))^2\right]\right]$$

Given $g^{(\mathcal{D})}(x) = a^{(\mathcal{D})}x$ and $\bar{g}(x) = \hat{a}x$, where $\hat{a} = \mathbb{E}_{\mathcal{D}}[a^{(\mathcal{D})}] = 1.43$, we have:

$$\text{variance} = \mathbb{E}_x \left[\mathbb{E}_{\mathcal{D}} \left[(a^{(\mathcal{D})}x - \hat{a}x)^2\right]\right]$$
$$= \mathbb{E}_x \left[x^2 \cdot \mathbb{E}_{\mathcal{D}} \left[(a^{(\mathcal{D})} - \hat{a})^2\right]\right]$$

These can be separated as one part is over x and the other over $\mathcal{D}$:

$$\text{variance} = \left(\mathbb{E}_x[x^2]\right) \cdot \left(\mathbb{E}_{\mathcal{D}} \left[(a^{(\mathcal{D})} - \hat{a})^2\right]\right)$$

$$= \left(\mathbb{E}_x[x^2]\right) \cdot \mathrm{Var}(a^{(\mathcal{D})})$$

- First Term: $\mathbb{E}_x[x^2]$ with $P(x) = \frac{1}{2}$ on $[-1, 1]$:

$$\mathbb{E}_x[x^2] = \int_{-1}^{1} x^2 \cdot \frac{1}{2}\, dx = \frac{1}{2}\left[\frac{x^3}{3}\right]_{-1}^{1} = \frac{1}{3}$$

- Second Term:

$$\mathrm{Var}(a^{(\mathcal{D})}) = \mathbb{E}[(a^{(\mathcal{D})})^2] - (\mathbb{E}[a^{(\mathcal{D})}])^2 = \mathbb{E}[(a^{(\mathcal{D})})^2] - (1.43)^2$$

  Given $\mathbb{E}[(a^{(\mathcal{D})})^2] \approx 2.70$,

$$\mathrm{Var}(a^{(\mathcal{D})}) = 2.70 - (1.43)^2 = 2.70 - 2.0449 = 0.6551$$

Substituting:

$$\begin{aligned} \text{variance} &= \frac{1}{3}\,(0.6551) \\ &= 0.218 \end{aligned}$$